Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/15/2005
PI Name:	Veretennikov, Igor N. Ph.D.		
Project Title:	Diffusive Coarsening of Liquid Foams in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	46556-5670	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	03/01/2000	**End Date:**	11/30/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Glaizer, James		
COI Name (Institution):	Glazier, James A. (Indiana University, Bloomington)		
Grant/Contract No.:	NAG3-2366		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This will include investigating how liquid redistributes throughout the foam microstructures as time evolves. Changes in the membrane thickness distribution in wet foams will be monitored with the foam analyzer. This invasive technique involves probing the foam by inserting a thin glass fiber at a constant speed and simultaneously and simultaneously measuring the light reflected from the tip as it passes through gas-liquid phase boundaries. Techniques adapted from image analysis will be used to infer bubble size distribution and membrane thickness. Optical tomography will be used to image the evolving microstructure of dry foams. This involves using a laser beam to illuminate foam lamellae and inverting the data to reconstruct the full foam structure including the bubble topologies and all correlations. It also proposes to use magnetic resonance imaging (MRI) to measure effective foam viscosity in ground-based experiments. Counter-current injection of liquid will be used to balance the gravity-driven drainage and maintain the liquid content of flowing foam. The objective of this proposal is to perform systematic ground-based experiments and in microgravity on diffusion-driven coarsening in a series of 3-D foams with different liquid content to test, the crossover from Von		

	Neumann to Lifschitz growth laws. The main focus of this proposal is to study the pure diffusion-driven aging of three-dimensional liquid foams. It has two major space related focus. The first one is on the fundamental science research on foams and secondly the making and engineering of high performance materials.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The main goal of this project is to better understand foam behavior both on Earth and in microgravity conditions and to determine the relation between a foam's structure and wetness and its rheological properties. To mimic foam behavior under microgravity, the foams are stabilized by supplying extra fluid on the top of the foam head. Such stabilized foam can stand up to few weeks while draining foam, even based on highly viscous fluids and with addition of stabilizers, survives 2-3 days maximum. We have showed, that the pure diffusive coarsening of stabilized foam is extremely slow if ruptures of membranes are negligible. Films ruptures not only result in much faster coarsening but also strongly affect bubble size distribution (BSD) leading to appearance of significant fraction of very large bubbles. Such foams are convectively unstable if the stabilizing flow rate exceeds some critical values. The foam stability to such convective instabilities depends strongly not only on the average bubble size distribution but on the spatial bubble distribution as well. The foam's meltdown occurs at significantly lower liquid fracture than predicted close packed transition. We have estimated this critical flow rate. In experiments with the liquid foams that undergo the gelation transition, we have found that non-uniform temperature profile across the foam cell and drainage can create thicker plateau borders forming "quasi-periodic" channels going from the wall to the center of the cell. Since such foam generation process is quite similar to that is used for solid foam insulation, we investigated the physical mechanism of this instability.
Bibliography Type:	Description: (Last Updated: 12/29/2020)
Articles in Peer-reviewed Journals	Vasconcelos IF, Cantat I, Glazier JA. "Dynamics and topological aspects of a reconstructed two-dimensional foam time series using Potts model on a pinned lattice." Journal of Computational Physics. 2003 Nov 20;192(1):1-20. https://doi.org/10.1016/j.jcp.2003.06.001 , Nov-2003